Université Grenoble Alpes
GIPSA-lab — Infinity Team

Internship Report

Modeling and Control of a PEM Fuel Cell Stack

From physical principles to per-cell calibration and dynamic characterization

Author:
Khalil Ellah Lagha

Supervisor:
Prof. Emmanuel Witrant

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Abstract

This report presents the modeling, calibration, and dynamic characterization of a proton exchange membrane fuel cell (PEMFC) stack carried out at GIPSA-lab. A reduced control-oriented model is developed from electrochemical principles: a Nernst open-circuit voltage combined with semi-empirical activation, ohmic, and concentration losses, coupled to lumped-volume hydrogen, membrane-hydration, and cathode-humidity dynamics. Six parameters per cell are identified from the Optimization Experiment using a two-stage optimization (differential evolution followed by L-BFGS-B), yielding root-mean-square voltage errors below 12 mV for all healthy cells and revealing degraded cells through voltage collapse and degenerate fits. The state-space matrices of the linearized model are written out explicitly — the state and input matrices follow directly from the dynamic equations, while the output matrices require a Jacobian linearization of the nonlinear voltage equation. The system is shown to be structurally controllable for all realistic electro-osmotic drag coefficients, while a voltage-only output observes the hydrogen pressure but not the water states. Step-load experiments (3.3 Ω and 6.8 Ω) characterize the current response of the stack ($\tau_I \approx$ 8–10 s), providing quantitative targets for closed-loop control design. Building on these results, a control architecture based on a DC/DC boost converter regulated by a PI controller is presented, in which the converter output is regulated to behave either as a current source — the approach of Derbeli et al. — or as a voltage source. All figures and detailed tables are gathered in the appendices.

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1 Introduction

Fuel cells convert the chemical energy of hydrogen directly into electricity, with water and heat as the only by-products. Among them, the proton exchange membrane fuel cell (PEMFC) is the leading candidate for transport and small stationary applications thanks to its low operating temperature, high power density, and fast start-up [9, 3]. Their efficient and safe operation, however, requires feedback control of the gas supply and load: insufficient reactant flow causes starvation and irreversible degradation, while excessive anode–cathode pressure difference can damage the membrane [16, 6]. Control design, in turn, requires a model that is simple enough for analysis and accurate enough to represent the actual hardware — including the differences between individual cells of a stack [1].

This internship, carried out at GIPSA-lab (Université Grenoble Alpes) under the supervision of Prof. Emmanuel Witrant, addresses this need for a small educational PEMFC stack of ten cells. The objectives are:

1. to develop a reduced, control-oriented dynamic model of a single cell and of the stack, grounded in the electrochemical literature;
2. to identify the model parameters *per cell* from experimental data acquired on the bench;
3. to derive the linearized state-space representation and analyze its structural controllability and observability, as a foundation for controller and observer design;
4. to characterize experimentally the dynamic response of the stack to resistive load steps, providing the timescales needed for control bandwidth selection;
5. to propose a PI-based control architecture for the PEMFC power system through a DC/DC boost converter, following [4].

The report is organized following a global vision of the work. Section 2 summarizes the operating principle of PEMFCs and positions the work in the literature. Section 3 presents the reduced model, the parameter identification procedure, the state-space analysis method, and the experimental protocols. Section 4 discusses the calibration, structural-analysis, and step-response results. Section 5 presents the control of the PEMFC power system with a DC/DC boost converter and a PI controller. Section 6 concludes and outlines perspectives. In accordance with the report rules, all figures are gathered in Appendix B and detailed derivations and tables in Appendices A and C; the main text is self-contained and readable without them.

2 Context & Background

2.1 Operating principle of a PEM fuel cell

A PEMFC converts the chemical energy of hydrogen and oxygen into electricity, heat, and water [1, 8, 12]. At the anode, hydrogen splits into protons and electrons; the protons cross the polymer membrane while the electrons supply the external load; at the cathode, oxygen recombines with both to form water (see Appendix B, Figure B.1). The overall reaction is

$$\mathrm{H_2} + \tfrac{1}{2}\mathrm{O_2} \rightarrow \mathrm{H_2O} + \mathrm{electricity} + \mathrm{heat}.$$

The membrane conducts protons, blocks electrons, and separates fuel from oxidant; together with the two electrodes it forms the membrane electrode assembly (MEA) [1, 5]. Water management is critical: a dry membrane loses proton conductivity, while excess water floods the gas paths; stable operation requires a balance between the two [14, 15].

2.2 Voltage losses and polarization behavior

A practical cell delivers less than the reversible voltage because of three loss mechanisms [1, 9, 11], dominant in different regions of the polarization curve (Appendix B, Figure B.2):

- **activation loss** — kinetic limitation of the electrode reactions, dominant at low current [9, 14];
- **ohmic loss** — resistive drop across membrane and conductors, approximately linear in current [9, 11];
- **concentration loss** — mass-transport limitation causing the sharp voltage drop at high current density [1, 14].

The fast transient behavior is additionally shaped by the charge double-layer effect at the electrode/electrolyte interface, which acts as a small capacitor in equivalent electrical models [9, 1].

2.3 State of the art

Control-oriented PEMFC modeling traces back to the system-level model of Pukrushpan [9], which combines a polarization-curve voltage model with compressor, manifold-filling, and membrane-hydration dynamics, and which most later models specialize or simplify [11, 7, 16]. Simplified electrical models reproduce the static V–I characteristic of small stacks with a handful of parameters [1, 10]; importantly, per-cell measurements on a small stack show voltage deviations of up to 7 % between cells [1], motivating the per-cell calibration adopted here. Parameter identification by direct-search optimization on measured voltage and power was formalized in [11], and modern metaheuristics extend it [2]. On the control side, PI regulation of flows, pressures, and converter currents is the established baseline [16, 4], with LQ multivariable control, time-delay control, and predictive schemes as documented upgrades [9, 7, 13]. At system level, the delays of the air and hydrogen supply components remain the limiting factor for dynamic operation [6]. A thematic synthesis of these references is maintained in the project documentation.

3 Methodology

3.1 Reduced control-oriented model

Following [11, 9], the cell is decomposed into three interacting parts — anode, membrane, and cathode — each lumped into a single control volume. The main assumptions are: hydrogen and air are the only reactants; channel pressure drops are neglected; gases follow the ideal-gas law; temperature is constant or slowly varying; and the oxygen partial pressure is fixed at $P^*_{O_2} = 0.21\, P_{\mathrm{atm}}$. The result is a semi-empirical reduced model: physically motivated mass balances combined with empirical loss relations that are calibrated from data [2].

Open-circuit voltage. The reversible voltage follows from the Gibbs free energy of the reaction and Faraday's law (full derivation in Appendix A):

$$E = 1.229 - 0.85\times10^{-3}(T_{fc} - 298.15) + 4.31\times10^{-5}\, T_{fc}\left[\ln p_{H_2} + \tfrac{1}{2}\ln p_{O_2}\right]\ \mathrm{V}, \tag{1}$$

with T_{fc} the stack temperature [K] and p_{H_2}, p_{O_2} the reactant partial pressures [atm] [9].

Voltage losses. The cell voltage is the reversible voltage minus three losses:

$$v_{fc} = E - v_{act} - v_{ohm} - v_{con}, \tag{2}$$

with the semi-empirical forms [9, 11, 2]

$$v_{act} = \xi_1 + \xi_2 T + \xi_3 T\ln(C_{O_2}) + \xi_4 T\ln(i_{FC}), \tag{3}$$

$$v_{ohm} = i_{FC}\Big(\rho_m \frac{t_m}{A_{fc}} + c\Big), \qquad \rho_m = \Big[(b_{11}\lambda_m - b_{12})\exp\big(b_2(\tfrac{1}{303} - \tfrac{1}{T_{fc}})\big)\Big]^{-1}, \tag{4}$$

$$v_{con} = -B\ln\Big(1 - \frac{i_{FC}}{i_{\max}}\Big), \tag{5}$$

where i_{FC} is the cell current, C_{O_2} the dissolved-oxygen concentration (Henry's law), t_m and A_{fc} the membrane thickness and active area, λ_m the membrane water content, and $\boldsymbol{\theta} = (\xi_1, \xi_2, \xi_3, \xi_4, B, c)$ the parameters identified per cell. The stack voltage is the sum over the n cells in series.

Anode dynamics. The hydrogen pressure follows a mass balance on the anode volume with Faraday consumption:

$$\dot P_{H_2} = \frac{R_{H_2}T}{V_{an}}\Big(W_{H_2,in} - W_{H_2,out} - \frac{M_{H_2}n}{2F}\, i_{FC}\Big), \tag{6}$$

and the anode water balance retains only the membrane flux, $\dot m_{w,an} = W_{v,mbr}$, with relative humidity $\phi_{an} = m_{w,an}R_vT/(V_{an}p_{sat}(T))$. P_{H_2} enters the Nernst term of (1) and is the primary control target.

Membrane and cathode. Water crosses the membrane mainly by electro-osmotic drag, $W_{v,mbr} = M_v A_{fc}\, n\, n_d\, i_{FC}/$ [9]; the membrane water content is given by the Springer sorption isotherm $\lambda(a) = 0.043 + 17.81a - 39.85a^2 + 36.0a^3$ averaged over the two faces. On the cathode side, the oxygen partial pressure is held constant at $P^*_{O_2} = 0.21\, P_{\mathrm{atm}}$, which removes the oxygen state while keeping the cathode water balance, $\dot{m}_{w,ca} = -W_{v,mbr} + W_{v,gen}$ with $W_{v,gen} = M_v n\, i_{FC}/(2F)$, whose humidity feeds back into the ohmic loss through λ_m. The block-level architecture of the resulting model is shown in Appendix B, Figure B.3. The model is compact and suitable for simulation and control design, at the price of ignoring spatial variations, oxygen transients, and detailed liquid-water effects [14, 15].

3.2 Per-cell parameter identification

The six parameters per cell are identified by minimizing the root-mean-square error (RMSE) between predicted and measured cell voltage over a calibration window:

$$\min_{\boldsymbol{\theta}} \sqrt{\frac{1}{N}\sum_{i=1}^{N}\big(v_{\mathrm{model}}(I_i, P_{H_2,i}; \boldsymbol{\theta}) - v_{\mathrm{meas},i}\big)^2}. \tag{7}$$

A two-stage strategy is used: **differential evolution** for the global search (population 20, up to 800 generations, fixed seed), followed by **L-BFGS-B** local refinement under the same box bounds. Physical constraints are imposed as bounds: $\xi_4 \leq 0$, $B \geq 0$, $c \geq 0$. The optimizer converges for most cells within a few tens of generations (convergence history in Appendix B, Figure B.4).

3.3 State-space analysis

Around an operating point $(\mathbf{x}_0, \mathbf{u}_0)$ the model is put in the standard linear state-space form $\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$, $y = C\mathbf{x} + D\mathbf{u}$. Controllability is assessed through the rank of $\mathcal{C} = [B\ AB\ A^2B]$ and observability through $\mathcal{O} = [C;\ CA;\ CA^2]$. The explicit construction of the four matrices — A and B read off the dynamic equations, C and D obtained by Jacobian linearization of the nonlinear voltage equation — is carried out in Section 4.2.

3.4 Experimental protocols

All experiments were performed on the ten-cell educational PEMFC stack of the laboratory bench, with per-cell voltage acquisition at 1 Hz through the serial data-acquisition chain developed in the project.

- **Optimization Experiment** (April 2026): a fixed $25\,\Omega$ resistor is connected at $t = 135$ s and removed at $t = 684$ s, with no valve-driven current control. The calibration points used for the per-cell parameter identification are drawn from the loaded window of this experiment.
- **TR-current step experiments** (9 June 2026): the stack is left at open circuit, then a fixed resistor ($3.3\,\Omega$ for Experiment 1, $6.8\,\Omega$ for Experiment 4) is connected at $t = 0$. The processing pipeline detects the connection event from a > 100 mA current jump and re-zeros the time axis. The current step response is fitted with the exponential model $I(t) = I_{\mathrm{ss}} + \Delta I\, e^{-t/\tau_I}$, from which settling times $T_{s,p\%} = -\tau_I \ln(p/100)$ are derived analytically; the 90 %→10 % fall time is measured on the raw signal.

4 Results & Analysis

4.1 Parameter identification on the Optimization Experiment

The two-stage optimizer identified the six semi-empirical parameters for all ten cells from the loaded window of the Optimization Experiment. For the healthy cells the calibration RMSE ranges from 2.0 to 11.3 mV — below 12 mV in all cases — and the mean time-series RMSE over the healthy cells is 8.0 mV. The per-cell V–I fits, the time-series comparison, the stack-level validation, and the error traces are shown in Appendix B, Figures B.5–B.8; the complete table of tuned parameters is given in Appendix C, Table C.1.

Two findings stand out. First, the model describes healthy cells accurately with only six parameters: the identified activation coefficients are consistent across cells, and the concentration (B) and contact-resistance (c) terms remain small, indicating that activation and ohmic effects dominate over the covered operating range. Second, the identification doubles as a *diagnosis* tool: Cells 0, 1, and 2 collapse to approximately 0 V under load and the optimizer returns identical, degenerate parameter sets for them (RMSE < 0.1 mV against a flat near-zero voltage) — a clear signature of hardware degradation rather than a physically meaningful fit, since no consistent parameter combination within the physical bounds can reproduce a non-functioning electrode.

4.2 State-space representation, controllability and observability

4.2.1 State-space representation of the fuel cell

The linearized model uses the state, input, and output vectors

$$\mathbf{x} = \begin{bmatrix} P_{H_2} \\ m_{w,an} \\ m_{w,ca} \end{bmatrix}, \qquad \mathbf{u} = \begin{bmatrix} i_{FC} \\ W_{H_2,in} \end{bmatrix}, \qquad y = v_{fc},$$

where P_{H_2} is the anode hydrogen partial pressure, $m_{w,an}$ and $m_{w,ca}$ the water masses stored on the anode and cathode sides, i_{FC} the load current (a disturbance imposed by the load), $W_{H_2,in}$ the hydrogen inlet flow (the manipulated input), and v_{fc} the measured cell voltage. In the standard form $\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$, $y = C\mathbf{x} + D\mathbf{u}$, the four matrices have a direct physical reading: A describes the internal dynamics (water exchange across the membrane; the hydrogen pressure behaves as an integrator), B describes how the current drawn and the inlet flow drive the states, C describes how the states appear in the measured voltage, and D is the direct feedthrough from the inputs to the voltage.

Matrices A and B — read directly from the dynamic equations. The three state equations of Section 3.1 are linear in the states and inputs once the temperature is held constant: the hydrogen pressure balance (6) (with $W_{H_2,out} \approx 0$ for the dead-end anode) and the two water balances written on the membrane water-transport terms,

$$\dot{m}_{w,an} = -\beta\, m_{w,an} + \gamma\, m_{w,ca} + d_1\, i_{FC}, \qquad \dot{m}_{w,ca} = \beta\, m_{w,an} - \gamma\, m_{w,ca} + d_2\, i_{FC},$$

with $\beta = D_w/(V_{an} t_m)$, $\gamma = D_w/(V_{ca} t_m)$, $d_1 = n_d M_v/F$, and $d_2 = (n - 2n_d)M_v/(2F)$. No linearization is needed; the matrices follow immediately:

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -\beta & \gamma \\ 0 & \beta & -\gamma \end{bmatrix}, \qquad B = \begin{bmatrix} -\dfrac{R_{H_2} T\, M_{H_2}\, n}{2F\, V_{an}} & \dfrac{R_{H_2} T}{V_{an}} \\ d_1 & 0 \\ d_2 & 0 \end{bmatrix}. \tag{8}$$

The first column of B is the effect of the load current (hydrogen consumption and electro-osmotic water displacement); the second column shows that the inlet flow acts on the hydrogen pressure only.

Matrices C and D — obtained by Jacobian linearization. Unlike the state equations, the output equation $y = v_{fc} = E(P_{H_2}) - v_{act}(i_{FC}) - v_{ohm}(i_{FC}, \lambda_m(m_{w,an}, m_{w,ca})) - v_{con}(i_{FC})$ is *nonlinear* in both states and inputs — it contains logarithms of P_{H_2} and i_{FC} and the nonlinear membrane-resistivity law (4) — so C and D cannot be read off the equations. They are obtained by a first-order Taylor (Jacobian) expansion around the operating point $(\mathbf{x}_0, \mathbf{u}_0)$:

$$\delta y \;\approx\; \underbrace{\left.\frac{\partial v_{fc}}{\partial \mathbf{x}}\right|_{(\mathbf{x}_0,\mathbf{u}_0)}}_{C} \delta\mathbf{x} \;+\; \underbrace{\left.\frac{\partial v_{fc}}{\partial \mathbf{u}}\right|_{(\mathbf{x}_0,\mathbf{u}_0)}}_{D} \delta\mathbf{u}.$$

The three steps are:

1. *Hydrogen-pressure entry.* Only the Nernst term (1) depends on P_{H_2}: $\partial v_{fc}/\partial P_{H_2} = 4.31\times 10^{-5}\, T/P_{H_2,0}$.
2. *Water-state entries.* The water masses act on the voltage through the chain $m_w \to \phi \to \lambda_m \to \sigma_m \to v_{ohm}$. Differentiating this chain gives

$$\frac{\partial v_{fc}}{\partial m_{w,an}} = \kappa_0\, \lambda'(\phi_{an,0})\, \frac{R_v T}{2V_{an} p_{sat}(T)}, \qquad \frac{\partial v_{fc}}{\partial m_{w,ca}} = \kappa_0\, \lambda'(\phi_{ca,0})\, \frac{R_v T}{2V_{ca} p_{sat}(T)},$$

with $\kappa_0 = i_0 t_m b_{11} \exp\!\left[b_2(\tfrac{1}{303} - \tfrac{1}{T})\right]/(A_{fc}\, \sigma_{m,0}^2)$ and $\lambda'(a) = 17.81 - 79.70a + 108a^2$ (derivative of the Springer isotherm).

3. *Input entries.* Differentiating the three loss terms with respect to i_{FC} at i_0, and noting that $W_{H_2,in}$ does not enter the output equation directly:

$$\frac{\partial v_{fc}}{\partial i_{FC}} = -\Big(\frac{\xi_4 T}{i_0} + \rho_{m,0}\frac{t_m}{A_{fc}} + c + \frac{B}{i_{\max} - i_0}\Big), \qquad \frac{\partial v_{fc}}{\partial W_{H_2,in}} = 0.$$

The resulting output matrices at the operating point are

$$C = \begin{bmatrix} \dfrac{4.31 \times 10^{-5}\, T}{P_{H_2,0}} & \kappa_0\, \lambda'(\phi_{an,0})\, \dfrac{R_v T}{2V_{an} p_{sat}(T)} & \kappa_0\, \lambda'(\phi_{ca,0})\, \dfrac{R_v T}{2V_{ca} p_{sat}(T)} \end{bmatrix}, \tag{9}$$

$$D = \begin{bmatrix} -\Big(\dfrac{\xi_4 T}{i_0} + \rho_{m,0}\dfrac{t_m}{A_{fc}} + c + \dfrac{B}{i_{\max} - i_0}\Big) & 0 \end{bmatrix}. \tag{10}$$

In the *simplified* model where λ_m is held constant, the two water-state entries of C vanish and $C_{\text{red}} = \begin{bmatrix} 4.31 \times 10^{-5}\, T/P_{H_2,0} & 0 & 0 \end{bmatrix}$.

4.2.2 Controllability

Because P_{H_2} is directly driven by $W_{H_2,in}$ (entry $R_{H_2}T/V_{an} > 0$ in B), controllability of the full system reduces to controllability of the two-state water subsystem (A_w, B_w) with $A_w = \left[\begin{smallmatrix} -\beta & \gamma \\ \beta & -\gamma \end{smallmatrix}\right]$ and $B_w = [d_1; d_2]$ from (8). Its controllability determinant, $\det(\mathcal{C}_w) = (d_1 + d_2)(\beta d_1 - \gamma d_2)$ with $d_1 + d_2 = nM_v/(2F) > 0$, vanishes only under the single degenerate condition

$$n_d = \frac{n\, V_{an}}{2\,(V_{an} + V_{ca})}, \tag{11}$$

i.e. when electro-osmotic drag exactly cancels the back-diffusion transport. **The system is therefore fully controllable for all physically realistic drag coefficients.**

4.2.3 Observability

With voltage as the only measurement, the observability matrix $\mathcal{O} = [C;\, CA;\, CA^2]$ is built from (9). In the simplified model the output reduces to C_{red}: **voltage observes the hydrogen pressure but not the two water states.** In the full model all three states are potentially observable, but the water-state sensitivities in (9) are small and degrade away from the operating point. The practical consequence is that hydrogen-pressure regulation can rely on a reduced-order voltage-based observer, whereas a full-state observer would require additional pressure or humidity sensors.

4.3 Step-response characterization (TR-current experiments)

In both step experiments, Cells 1 and 2 read approximately 0 mV throughout the loaded window, confirming the degradation first observed in the Optimization Experiment; all other cells behave normally. The measured channels and the annotated step responses are shown in Appendix B, Figures B.9 and B.10. Table 1 summarizes the metrics.

Table 1: Response-time metrics for the current step response.

Current metric	**Exp 1 (3.3 Ω)**	**Exp 4 (6.8 Ω)**
I_{ss} [mA]	572.9	556.1
ΔI [mA]	82.1	52.9
τ_I [s]	8.2	10.3
$T_{s,2\%}$ [s]	32.0	40.2
Fall time 90%→10% [s]	16.0	19.0

The current time constant is about 25 % longer under the lighter load ($\tau_I = 10.3$ s at 6.8 Ω vs. 8.2 s at 3.3 Ω), consistent with slower electrochemical dynamics at lower current density; the initial perturbation ΔI is also smaller. The settling times derived from the exponential fit ($T_{s,2\%} = 32$–40 s) are robust against measurement noise, since they are computed analytically from the fitted τ_I rather than from the noisy raw signal. This timescale captures the dominant dynamics seen by the load — electrochemical kinetics combined with the gas-supply transients — and provides a quantitative target for controller bandwidth selection: the closed-loop control of Section 5 must act well within $\tau_I \approx 8$–10 s to shape the stack's transient behavior.

5 Control of the PEMFC Power System: DC/DC Boost Converter with PI Controller

This section presents the control architecture adopted for the PEMFC power system, following the reference article of Derbeli et al. [4].

5.1 System overview

By itself, a PEMFC stack is *neither* an ideal voltage source *nor* an ideal current source: its terminal voltage V_{STACK} is low and varies strongly with the current drawn, following the polarization curve of Section 2. The purpose of the control stage is to shape this raw output into a *regulated source*. To this end the stack feeds a **DC/DC boost converter** (boost chopper), and a **PI controller** closes the loop by acting on the converter duty cycle T through pulse-width modulation (PWM) [4]. The block diagram of the architecture is shown in Appendix B, Figure B.11.

Depending on which variable the PI loop regulates, the *converter output* — i.e. the PEMFC power system seen from the load — can be made to behave as:

- a **controlled current source** — the loop regulates the inductor (stack) current I_L to a reference I_{REF}; the approach used in the reference article, detailed in Section 5.2;
- a **controlled voltage source** — the loop regulates the output voltage V_{OUT} to a reference V_{REF}; an alternative covered in Section 5.3.

It is therefore the regulated converter output, not the fuel cell itself, that is made to act as a current or a voltage source.

5.2 Output regulated as a current source (article approach)

Here the goal is to make the regulated converter output behave as a **current source**. Following [4], the control design proceeds in five steps.

1. *Control objective.* The controlled variable is the inductor (stack) current I_L, which the PI loop forces to track a reference I_{REF} so that the system delivers a regulated current; in the article, $I_{REF} = 9.74\,\mathrm{A}$ corresponds to the maximum-power operating point of the 50 W, ten-cell stack used for validation (54 W at 5.54 V). Regulating the current directly also protects the stack, since its maximum current (10 A) must never be exceeded.

2. *Boost converter state-space model.* Averaging the two switching topologies of the converter (switch ON / switch OFF) over one PWM period yields, with state $x = [\, i_L \;\; V_{OUT} \,]^\top$, duty cycle T, load R, inductance L, and capacitance C [4]:

$$\dot{x} = \begin{bmatrix} 0 & \dfrac{T-1}{L} \\ \dfrac{1-T}{C} & -\dfrac{1}{RC} \end{bmatrix} x + \begin{bmatrix} \dfrac{1}{L} \\ 0 \end{bmatrix} u, \qquad y = \begin{bmatrix} 0 & 1 \end{bmatrix} x. \tag{12}$$

3. *PI control law.* The error signal is $e = I_{REF} - I_L$, and the control output (duty cycle) combines proportional and integral action:

$$u = K_P\, e + K_I \int e\, dt. \tag{13}$$

   The proportional term shapes rise time and the integral term removes the steady-state error, at the cost of possible overshoot if the gains are too aggressive.

4. *PI tuning (online method).* The gains are tuned experimentally, without knowledge of the plant transfer function [4]: set $K_P = 0$ and $T_i = \infty$; increase K_P until sustained oscillations appear and note this value K_{PU}; set $K_P = K_{PU}/2$; then decrease T_i until oscillations reappear (T_{iU}) and set $T_i = 2\,T_{iU}$.

5. *Expected behavior.* With properly tuned gains, the PI controller produces a smooth and stable rise of I_L to I_{REF} ($\approx 400\,\mathrm{ms}$ in the article) and good disturbance rejection even under large load variations, as validated experimentally on a dSPACE 1104 real-time platform [4].

5.3 Output regulated as a voltage source (alternative approach)

Here the goal is to make the same regulated converter output behave instead as a **voltage source**, using the same PI-based methodology.

1. *Control objective.* The controlled variable is now the converter output voltage V_{OUT}, which the PI loop forces to track a reference V_{REF} so that the system delivers a regulated voltage to the load.
2. *Boost converter model.* The same averaged state-space model (12) applies; the output of interest is the second state, $y = V_{OUT}$.
3. *PI control law.* The error signal becomes $e = V_{REF} - V_{OUT}$, and the same control law (13) applies.
4. *Tuning.* The same online tuning procedure can be used, adapted to the voltage-loop dynamics (which include the RC output stage and are therefore slower than the current loop).
5. *Comparison of the two objectives.* Regulating the current (the article's approach) is better suited for protecting the PEMFC from overcurrent, since the controlled variable is directly the current drawn from the stack; regulating the output voltage is more natural for supplying a fixed voltage to the load. Both make the converter output behave as a regulated source, rely on the same PI structure and converter model, and differ only in the feedback variable (I_L vs. V_{OUT}) and in the physical meaning of the reference signal.

6 Conclusion & Perspectives

This work developed a reduced control-oriented model of a ten-cell PEMFC stack, calibrated it per cell on the Optimization Experiment, derived and analyzed its linearized state-space representation, characterized its dynamic current response to load steps, and proposed a PI-based control architecture for the PEMFC power system.

The main outcomes are: (i) a six-parameter semi-empirical voltage model coupled to lumped anode, membrane, and cathode dynamics that fits all healthy cells to within 12 mV; (ii) a diagnosis capability — degraded cells (0, 1, 2) are revealed both by voltage collapse and by degenerate, physically meaningless fits; (iii) an explicit state-space representation in which A and B follow directly from the dynamic equations while C and D require Jacobian linearization of the nonlinear voltage equation, leading to proof that the model is generically controllable while a voltage-only observer can reconstruct the hydrogen pressure but not the water states; (iv) an experimental characterization of the current response to load steps ($\tau_I \approx 8$–$10\,\mathrm{s}$, $T_{s,2\%} \approx 32$–$40\,\mathrm{s}$); and (v) a control architecture based on a DC/DC boost converter regulated by a PI controller [4], in which the converter output is regulated to behave either as a current source (overcurrent protection, the article's approach) or as a voltage source (output-voltage regulation).

Perspectives follow directly. On the control side, the next step is to implement the proposed boost-converter PI loop on the bench and tune it online following [4], with the identified current timescale bounding the required closed-loop bandwidth; the observability analysis further supports a hydrogen-pressure regulator with a reduced-order voltage-based observer, with PI control as the literature-standard baseline [16, 4] and LQ or time-delay control as upgrades [9, 7]. On the modeling side, capturing oxygen transients (removing the constant-P_{O_2} assumption), liquid-water accumulation, and gas-supply delays [6] would extend validity at higher dynamics. Finally, replacing the degraded cells and repeating the calibration would allow stack-level closed-loop experiments.

References

[1] A. E. Andrea, M. Mañana, A. Ortiz, C. Renedo, L. I. Eguíluz, S. Pérez, and F. Delgado. A simplified electrical model of small pem fuel cell. *Renewable Energy and Power Quality Journal*, 1(4), 2006.

[2] Yan Cao, Yiqing Li, Geng Zhang, Kittisak Jermsittiparsert, and Navid Razmjooy. Experimental modeling of pem fuel cells using a new improved seagull optimization algorithm. *Energy Reports*, 5:1616–1625, 2019.

[3] WRW Daud, RE Rosli, EH Majlan, SAA Hamid, R Mohamed, and T Husaini. Pem fuel cell system control: A review. *Renewable Energy*, 113:620–638, 2017.

[4] Mohamed Derbeli, Maissa Farhat, Oscar Barambones, and Lassaad Sbita. Control of proton exchange membrane fuel cell (PEMFC) power system using PI controller. In *Proceedings of the International Conference on Green Energy Conversion Systems*. IEEE, 2017.

[5] V. Ionescu. Water and hydrogen transport modelling through the membrane-electrode assembly of a pem fuel cell. *Physica Scripta*, 95:034006, 2020.

[6] Jamil Kharrat and Jan Haußmann. Influence of the gas supply component delay on the dynamic operation of the PEM fuel cell system. In *Proceedings of the Hydrogen Technology Conference*, 2025.

[7] Y. B. Kim and S. J. Kang. Time delay control for fuel cells with bidirectional dc/dc converter and battery. *International Journal of Hydrogen Energy*, 35:8792–8803, 2010.

[8] A. K. M. Mohiuddin, N. Basran, and A. A. Khan. Modelling and validation of proton exchange membrane fuel cell (pemfc). *IOP Conference Series: Materials Science and Engineering*, 290:012026, 2018.

[9] Jay T. Pukrushpan. *Modeling and Control of Fuel Cell Systems and Fuel Processors*. PhD thesis, The University of Michigan, 2003.

[10] M. Shen and K. Scott. Power loss and its effect on fuel cell performance. *Journal of Power Sources*, 148:24–31, 2005.

[11] K. K. T. Thanapalan, J. G. Williams, G. P. Liu, and D. Rees. Modelling of a pem fuel cell system. In *Proceedings of the 17th IFAC World Congress*, 2008.

[12] A. Thomya and Y. Khunatorn. Design of control system of hydrogen and oxygen flow rate for proton exchange membrane fuel cell using fuzzy logic controller. In *Energy Procedia*, volume 9, pages 186–197, 2011.

[13] Yuhang Wang, Haotian Li, Huimin Feng, Kuihua Han, Suoying He, and Ming Gao. Simulation study on the pemfc oxygen starvation based on the coupling algorithm of model predictive control and pid. *Energy Conversion and Management*, 249:114851, 2021.

[14] Adam Z. Weber and John Newman. Modeling transport in polymer-electrolyte fuel cells. *Chemical Reviews*, 104(10):4679–4726, 2004.

[15] G. Zhang, L. Fan, J. Sun, and K. Jiao. A 3d model of pemfc considering detailed multiphase flow and anisotropic transport properties. *International Journal of Heat and Mass Transfer*, 115:714–724, 2017.

[16] Yun Zhu, Jianxiao Zou, Chao Peng, Yucen Xie, and Liying Li. Modelling and fuel flow control of pemfc considering over-pressure case. In *Proceedings of the IEEE Conference on Industrial Electronics and Applications*. IEEE, 2017.

A Derivation of the Open-Circuit Voltage

The global hydrogen–oxygen reaction is $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$. At non-standard conditions the Gibbs free energy change is

$$\Delta g_f = \Delta g_f^0 - \bar{R}T_{fc} \ln\left(\frac{p_{H_2}\, p_{O_2}^{1/2}}{p_{H_2O}}\right),$$

and for a reversible cell $\Delta g_f = -2FE$, so

$$E = -\frac{\Delta g_f^0}{2F} + \frac{\bar{R}T_{fc}}{2F} \ln\left(\frac{p_{H_2}\, p_{O_2}^{1/2}}{p_{H_2O}}\right).$$

Standard voltage. Taking liquid water as the product at 25 °C:

$$E^0 = -\frac{\Delta g_f^0}{2F} = -\frac{-237{,}200}{2 \times 96{,}485} \approx 1.229 \text{ V}.$$

Table A.1 lists Δg_f^0 at several temperatures.

Table A.1: Gibbs free energy change of the hydrogen reaction at various temperatures [9].

Product	**Temperature (°C)**	$\boldsymbol{\Delta g_f^0}$ **(kJ/mol)**
Liquid	25	−237.2
Liquid	80	−228.2
Gas	80	−226.1
Gas	100	−225.2
Gas	200	−220.4
Gas	600	−199.6
Gas	1000	−177.4

Temperature coefficient. From $\Delta G^0 = \Delta H^0 - T\Delta S^0$, with $\Delta S^0 = 69.95 - 130.7 - 102.58 \approx -163.3\ \mathrm{J\,mol^{-1}K^{-1}}$ for the liquid-water reaction,

$$\frac{\Delta S^0}{2F} \approx \frac{-163.3}{192{,}970} \approx -0.85 \times 10^{-3}\ \mathrm{V\,K^{-1}}.$$

Nernst pressure correction. Assuming $p_{H_2O} = 1$ (liquid product):

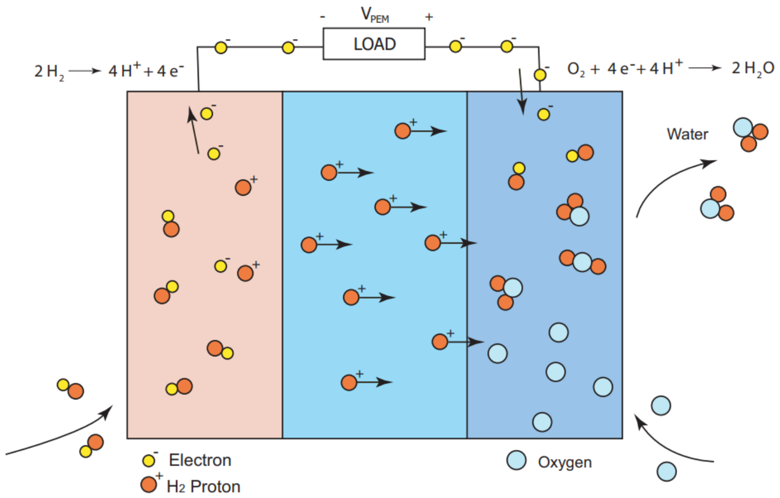
$$\frac{\bar{R}T_{fc}}{2F} \ln\!\left(p_{H_2}\, p_{O_2}^{1/2}\right) = \frac{8.314\, T_{fc}}{192{,}970} \left[\ln p_{H_2} + \tfrac{1}{2} \ln p_{O_2}\right] \approx 4.31 \times 10^{-5}\, T_{fc} \left[\ln p_{H_2} + \tfrac{1}{2} \ln p_{O_2}\right].$$

Final expression. Combining all terms yields Equation (1) of the main text.

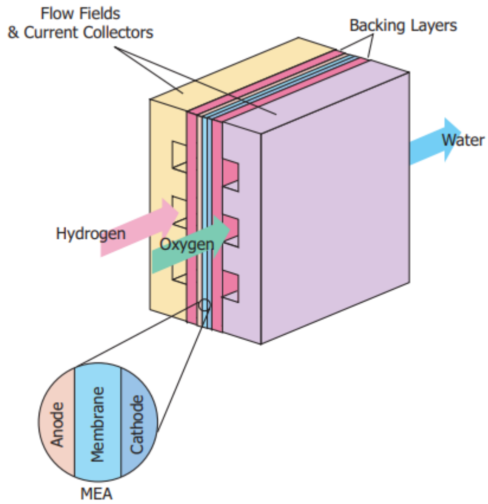
B Figures

All figures referenced in the main text are gathered here.

B.1 — Context & Background

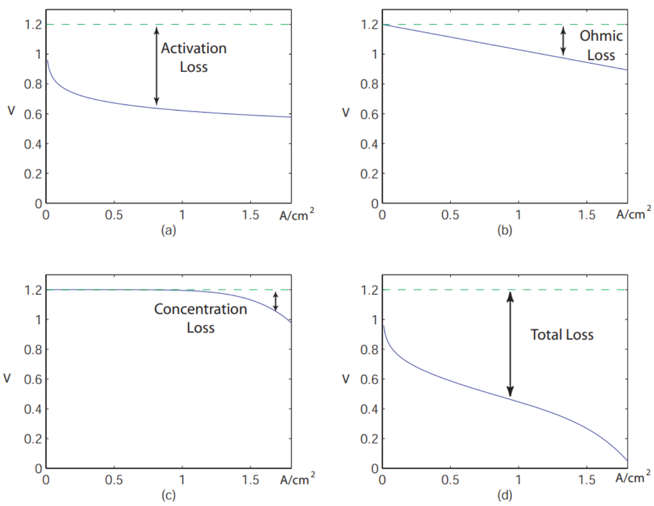


(a) Basic electrochemical operation [8, 12].

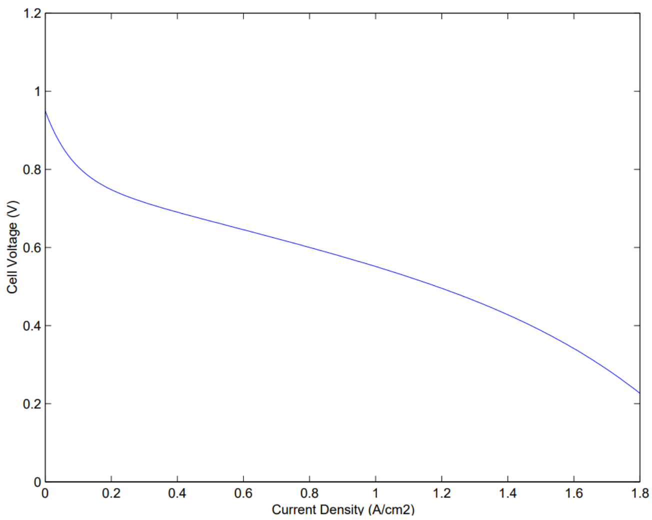


(b) Simplified stack structure and MEA [1].

Figure B.1: Operating principle and physical structure of a PEM fuel cell.



(a) Activation, ohmic, concentration, and total losses.



(b) Typical polarization curve.

Figure B.2: Influence of the main voltage losses on PEMFC performance.

B.2 — Model architecture

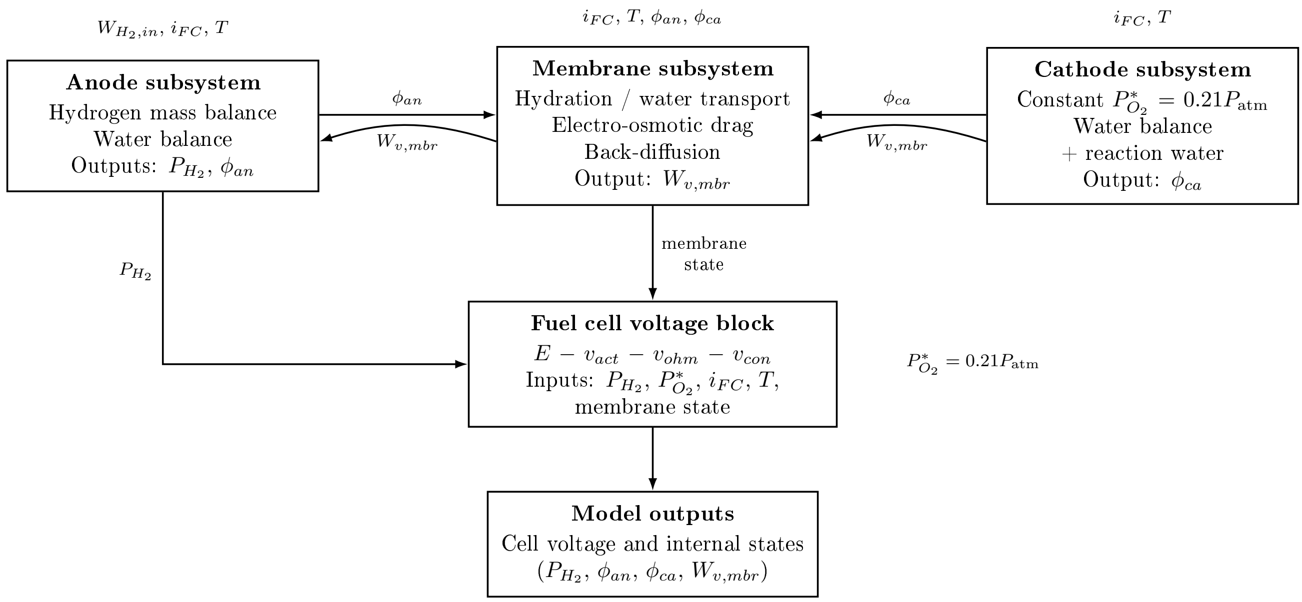


Figure B.3: Block-level architecture of the reduced PEMFC model.

B.3 — Parameter identification, Optimization Experiment

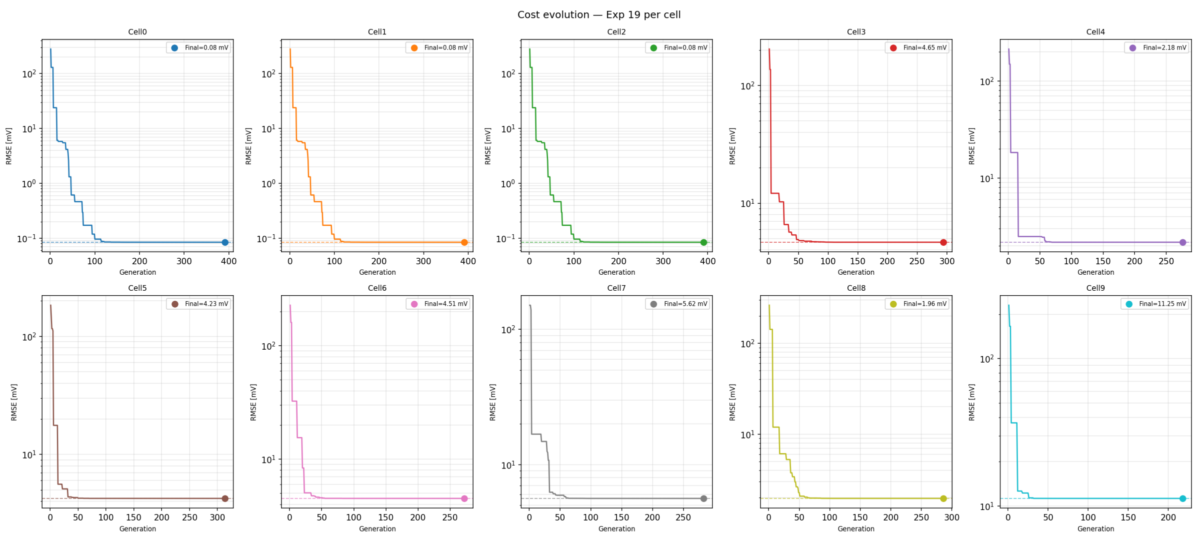


Figure B.4: DE optimizer convergence history for all ten cells — Optimization Experiment. Dashed line: final RMSE after L-BFGS-B refinement.

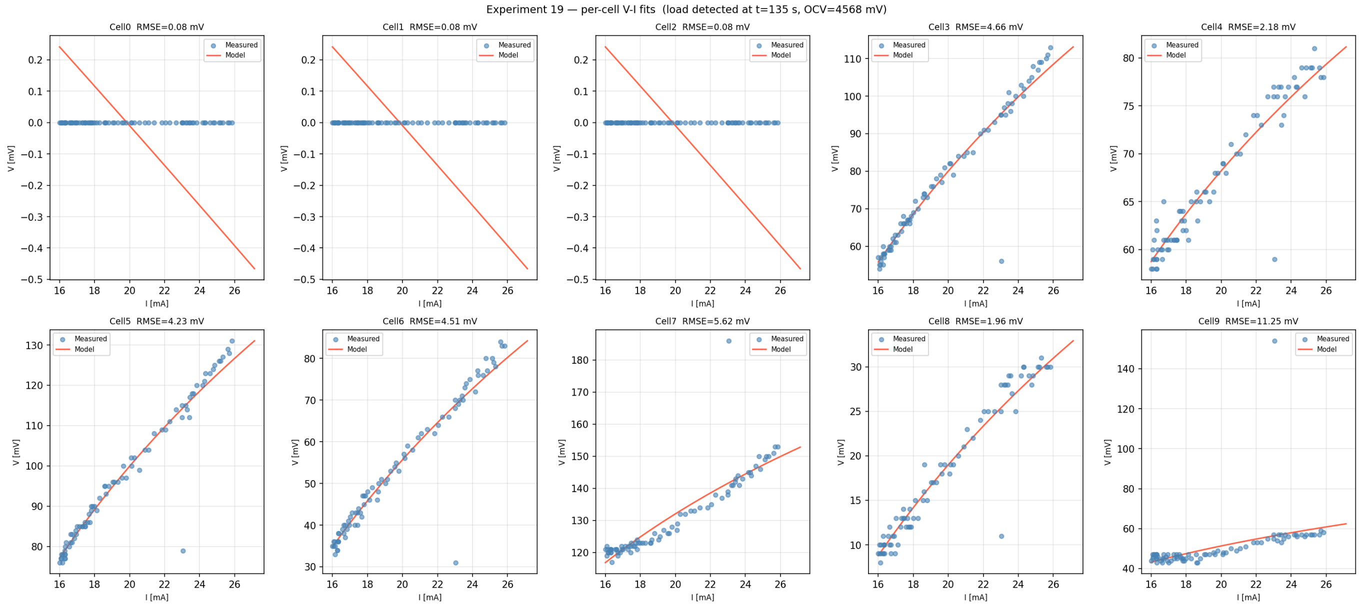


Figure B.5: Per-cell V–I calibration fits — Optimization Experiment.

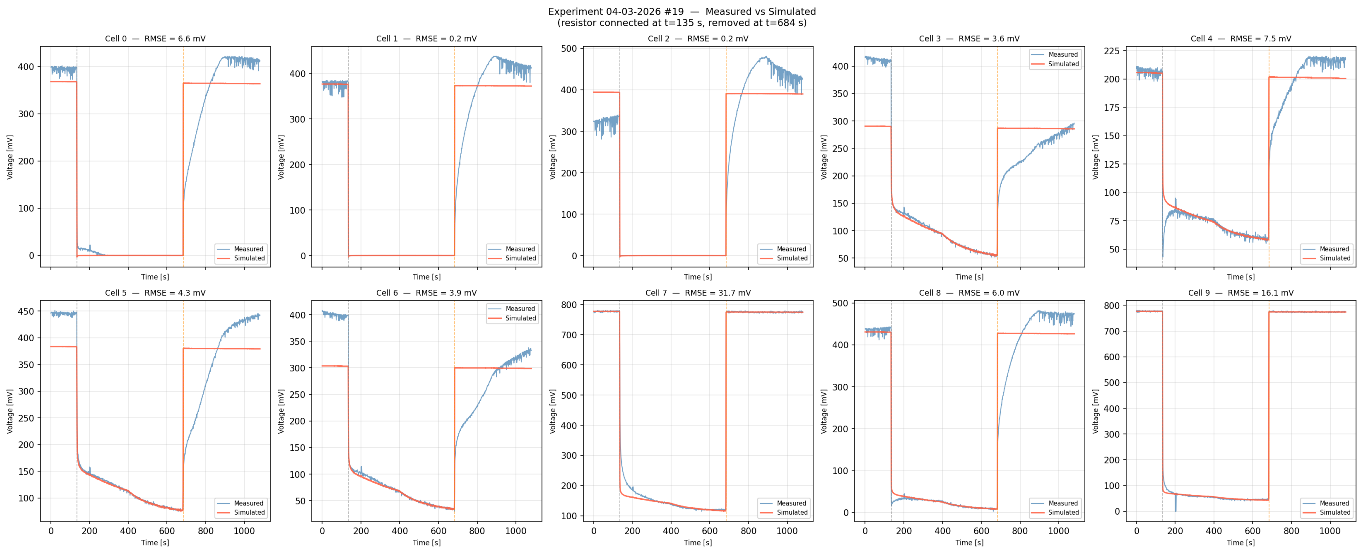


Figure B.6: Per-cell voltage, measured vs. simulated — Optimization Experiment. Cells 0, 1, 2 collapse under load (hardware degradation). Mean RMSE (healthy cells) = 8.0 mV.

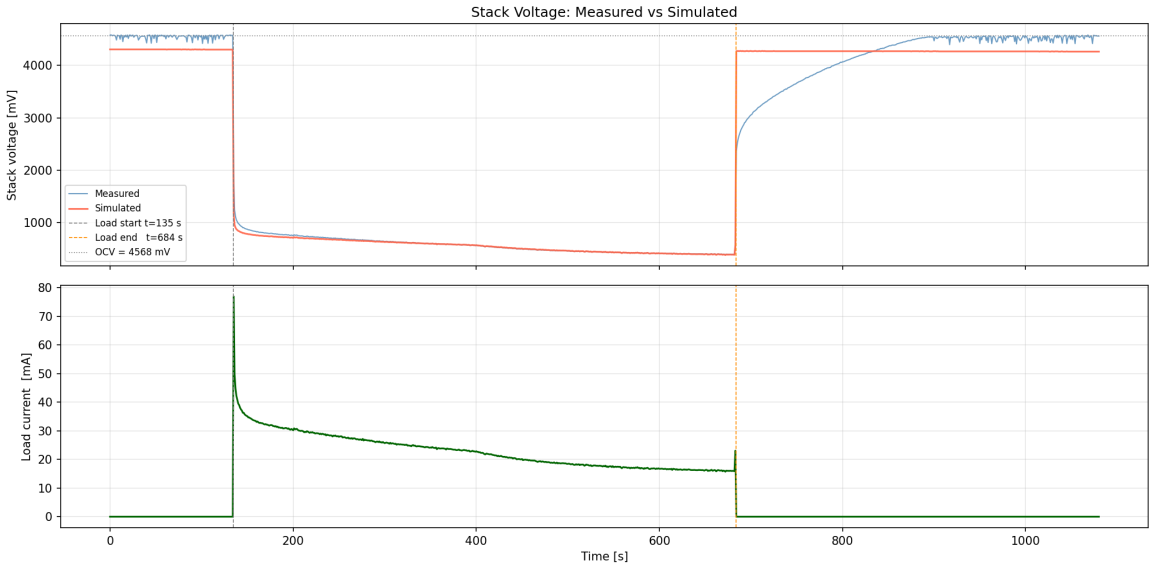


Figure B.7: Stack voltage, load current and hydrogen pressure — Optimization Experiment. Load applied at $t = 135\,\mathrm{s}$, removed at $t = 684\,\mathrm{s}$.

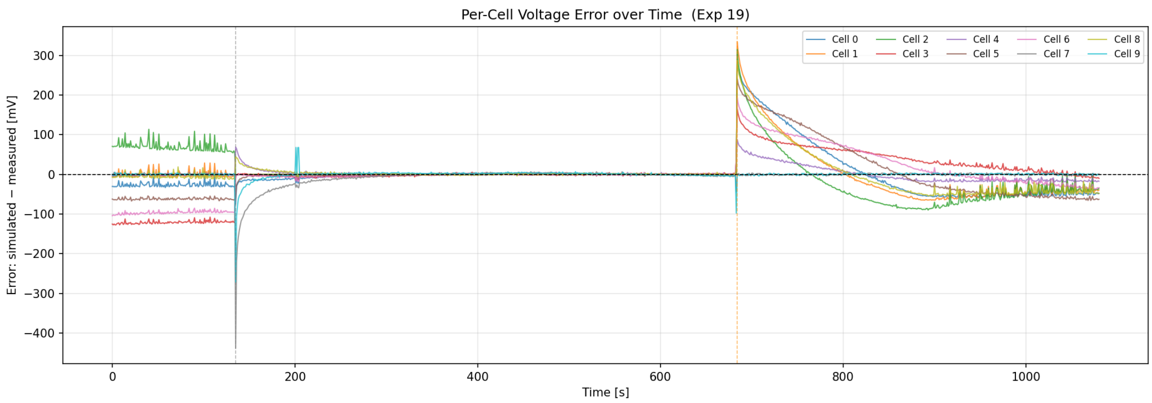
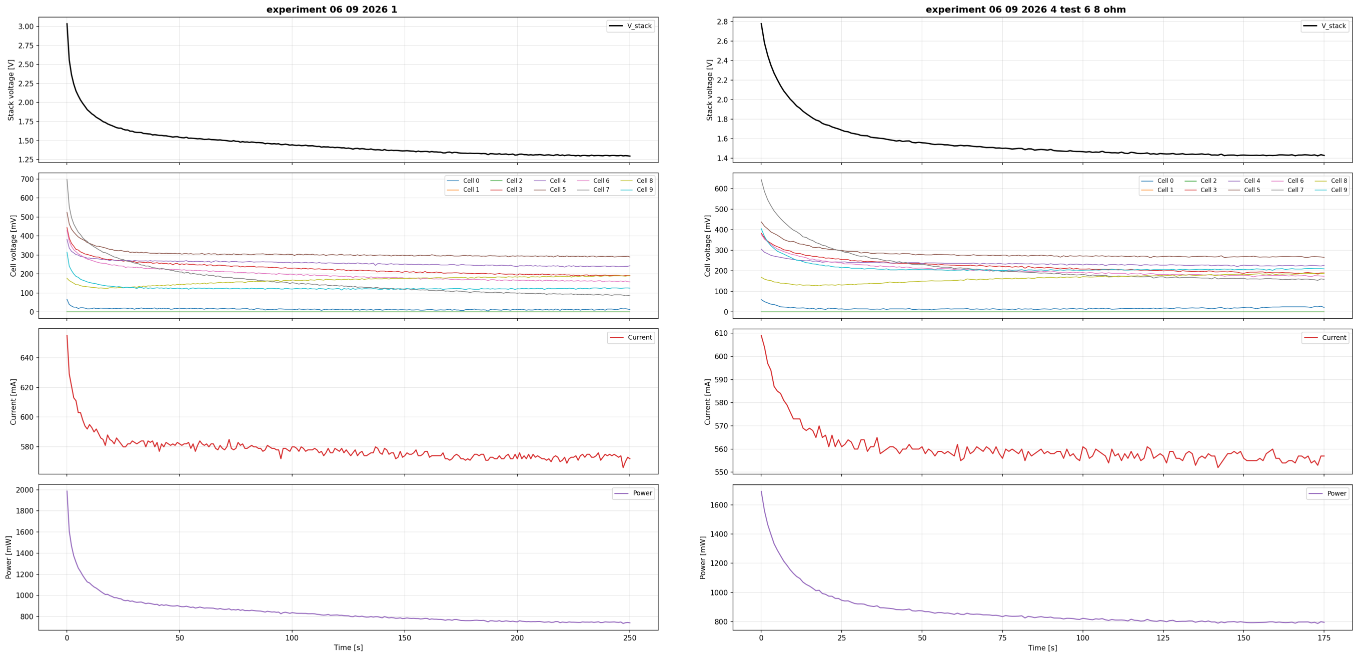


Figure B.8: Per-cell voltage error (simulated − measured) — Optimization Experiment.

B.4 — TR-current step-response experiments



(a) Experiment 1 ($R = 3.3\,\Omega$, 0–250 s).

(b) Experiment 4 ($R = 6.8\,\Omega$, 0–175 s).

Figure B.9: TR-current experiments: stack voltage [V], cell voltages [mV], current [mA], and power $P = V_{\text{stack}} \cdot I$ [mW].

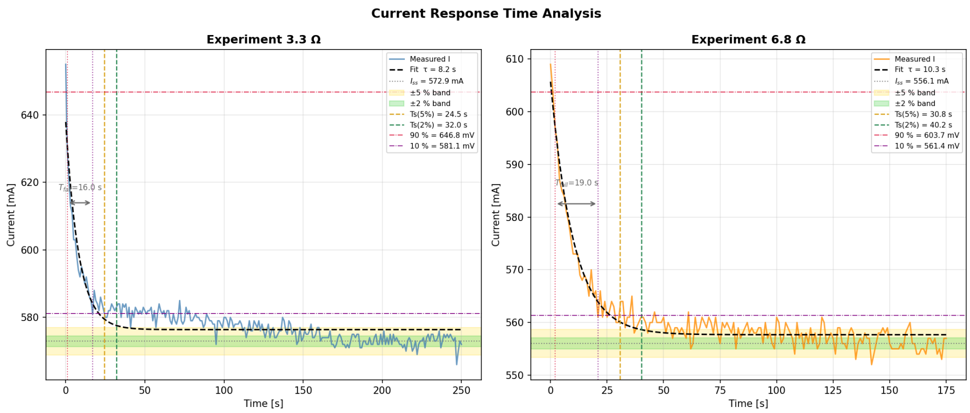


Figure B.10: Current step response after resistor connection: measured data (color), exponential fit (dashed), steady-state band, and settling-time markers.

B.5 — Control architecture

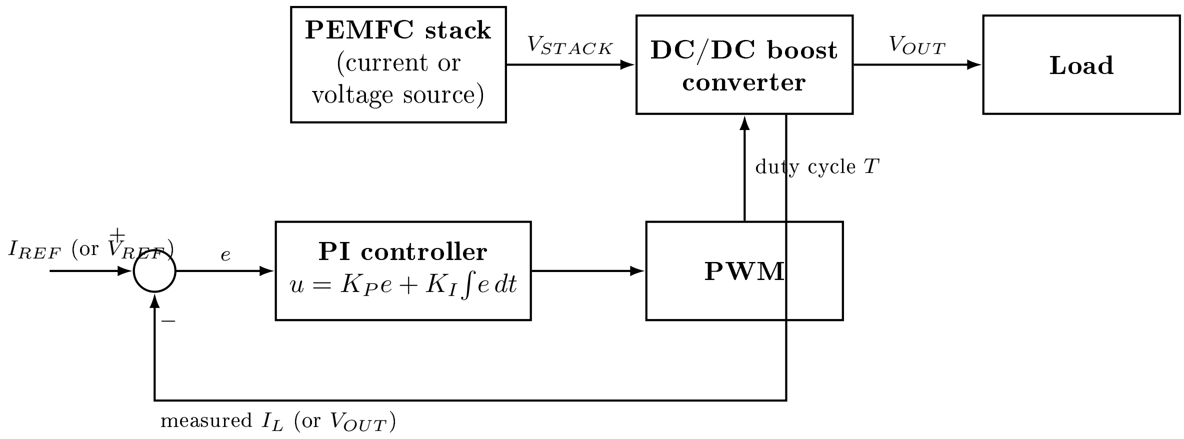


Figure B.11: Control architecture of the PEMFC power system, after [4]: the stack feeds a DC/DC boost converter whose duty cycle is set by a PI controller. Regulating the inductor current I_L makes the converter output behave as a current source; regulating the output voltage V_{OUT} makes it behave as a voltage source.

C Detailed Tables

Table C.1: Tuned parameters and calibration RMSE for all ten cells — Optimization Experiment. Cells 0–2 collapsed under load: the optimizer returns identical degenerate parameter sets that reproduce a near-zero voltage and are not physically meaningful.

Cell	RMSE [mV]	ξ_1	ξ_2	ξ_3	ξ_4	B	c
Cell 0†	0.08	-7.783	1.67×10^{-2}	-6.06×10^{-4}	≈ 0	1.08×10^{-2}	1.00×10^{-2}
Cell 1†	0.08	-7.783	1.67×10^{-2}	-6.06×10^{-4}	≈ 0	1.08×10^{-2}	1.00×10^{-2}
Cell 2†	0.08	-7.783	1.67×10^{-2}	-6.06×10^{-4}	≈ 0	1.08×10^{-2}	1.00×10^{-2}
Cell 3	4.66	-1.977	2.83×10^{-2}	1.32×10^{-3}	-3.17×10^{-4}	≈ 0	≈ 0
Cell 4	2.18	0.386	1.43×10^{-2}	8.11×10^{-4}	-1.24×10^{-4}	≈ 0	≈ 0
Cell 5	4.23	0.682	3.00×10^{-2}	1.93×10^{-3}	-2.98×10^{-4}	≈ 0	≈ 0
Cell 6	4.51	1.775	4.62×10^{-3}	4.87×10^{-4}	-2.73×10^{-4}	≈ 0	≈ 0
Cell 7	5.62	1.796	2.54×10^{-2}	1.82×10^{-3}	-1.99×10^{-4}	≈ 0	≈ 0
Cell 8	1.96	-4.946	2.81×10^{-2}	6.92×10^{-4}	-1.34×10^{-4}	≈ 0	≈ 0
Cell 9	11.25	-5.400	2.01×10^{-2}	9.31×10^{-5}	-1.06×10^{-4}	3.21×10^{-7}	≈ 0

† Degraded cell (voltage collapse under load); fit not physically meaningful.